Yijie Ma

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Skills

C/C++, Go, Python, Networking, Distributed Systems, Kubernetes, OpenTelemetry, Storage, Desktop App Development

Work Experience

Google — April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Architected C++ OpenTelemetry distributed tracing across HTTP and gRPC for Cloud, GenAI, and MCP APIs.
- Led Agent-to-Agent (A2A) observability and W3C Trace Context propagation, mentoring junior engineers.
- Spearheaded a decoupled API configuration rollout pipeline with dual-stack routing for zero-downtime delivery.
- Automated release risk analysis and staging promotions via LLM multi-agent workflows, cutting hours of manual toil.

Google Kubernetes Engine (GKE) Networking

- Engineered GKE control-plane automation in Go to auto-provision multi-NIC RDMA network topologies for AI clusters.
- Designed concurrent provisioning pipelines and validated multi-node GPUDirect-RDMA on NVIDIA H200 with NCCL.

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer (150+ PRs, 40K+ stars); architected the cross-platform EventEngine DNS resolver.
- Designed and built the C++ template Metrics framework and OpenTelemetry plugin, cutting counter latency by 10%.
- Co-authored gRFC A79: Non-per-call Metrics Architecture to standardize telemetry collection across gRPC core.
- Resolved critical concurrency bugs, security vulnerabilities, and performance bottlenecks for millions of OSS users.

Google Drive Desktop Application

- Engineered rsync-style differential upload and Gzip compression pipelines to cut upload bandwidth for millions of users.
- Developed the USB device backup stack processing millions of files daily and optimized I/O buffering for a 10x speedup.
- Designed and launched the cross-platform Preferences framework for user and enterprise admin policy configuration.
- Delivered 100+ crash fixes, memory optimizations, and reliability improvements across Windows, macOS, and Linux.

Stadia Cloud Gaming Service

- Accelerated game container click-to-play startup by 75% (1.58s to 396ms) by profiling and eliminating VM bottlenecks.
- Developed a Linux Cgroup management library to isolate and configure container resources during VM provisioning.
- Engineered a custom Windows shell replacing `explorer.exe` for gamer surface lockdown and process lifecycle control.

NetApp — August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Built a filesystem software and used RocksDB for cataloging users' snapshot data. Built full or incremental backup and restore functionalities.

Education

Carnegie Mellon University — Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University — Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)